US Accidents Severity

Data Preparation, Exploration and Feature Engineering

Data Science Midterm Project · Written Report
CRISP-DM phases 1-6: from understanding the problem to building features and planning the next steps

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Date: October 2026
Dataset: US Accidents (2016-2023), Moosavi et al. [1-3], a sample of 500,000 accidents
Code: notebooks 01-04, src/features.py and run_all.py in the project repository

Summary

Traffic centres need to know fast which accidents will cause long delays. In this project we prepare 500,000 accidents from the US Accidents dataset. Our goal is to predict the Severity of an accident (1 = small delay, 4 = long delay). We only use information that is known when the accident is first reported.

Here is what we did. We removed 11 columns that would give away the answer. We cleaned the data, and 495,532 accidents are left. We made 14 charts and ran 6 statistical tests. Then we built 36 inputs for a future model.

Here is what we found. Most accidents (80%) are Severity 2, so the classes are very unbalanced. The data provider (the company that recorded the accident) has the strongest link with Severity (Cramér's V = 0.29). This means the label partly depends on who recorded it. Accidents on highways are 2.5 times more often high-impact. Night and bad weather only matter a little. We have not trained a model yet. Section 6 explains our plan for that.

1. Business Understanding

1.1 Context

There are millions of road accidents in the US every year. Some only slow traffic for a few minutes. Others close a highway for hours. Right after an accident is reported, traffic centres and navigation apps must decide how many people to send and whether to send drivers on another road. A tool that guesses the impact of an accident from the first report would help them act faster.

1.2 Problem statement

Our question: if we know when, where and in what conditions an accident happens, can we predict how much it will slow down traffic (Severity 1-4)? We may only use information that is available at the moment the accident is reported.

Important: in this dataset, Severity means the effect on traffic (the delay). It does not mean how badly people were hurt. This is a classification problem with four classes, and the classes are very unbalanced.

1.3 Research questions

Table 1. Research questions. We answer each question with charts and, when possible, with a statistical test.

#	Research question	Answered in
RQ1	When do high-impact accidents happen (hour, weekday, day or night)?	4.3, 4.5
RQ2	Does the weather change how severe accidents are?	4.3, 4.5
RQ3	Does the type of road (highway, traffic lights, junctions) change severity?	4.3, 4.5, 5.1
RQ4	Is the Severity label the same across data providers and years?	4.4, 4.5
RQ5	Which information known at report time is most linked to Severity, and should go into a model?	5.3

1.4 Success criteria

For this midterm (preparing the data), we set five goals:

- **No leakage:** we only keep columns that are known when the accident is reported. We write down the reason for every column we remove.
- **Clear cleaning:** we write down the number of rows and columns before and after each step.
- **Evidence for each question:** a chart with labels and, when possible, a statistical test. We also report the effect size, not only the p-value.
- **Good reasons for each feature:** we choose the final features with clear rules, using only the training data.
- **Easy to repeat:** one command (python run_all.py) rebuilds every result. The random seed is fixed (42).

For the future model: a model that always says "Severity 2" is already right about 80% of the time. So accuracy is not a good score here. Instead we will use **macro F1** (the average F1 score over the 4 classes). It must be clearly higher than 0.22, which is the score of that lazy model. We will also look at the **recall of Severity 4** (how many of the worst accidents the model finds). It must be clearly above 0.

2. Data Understanding

2.1 Source

We use the US Accidents dataset made by Sobhan Moosavi and his team at Ohio State University [1-4]. It is on Kaggle under the CC BY-NC-SA 4.0 licence, which allows academic use if we cite it. The full dataset has about 7.7 million accidents from 49 US states, from February 2016 to March 2023.

The data was collected live from traffic APIs. There are three data providers, called Source1, Source2 and Source3. They share traffic events reported by transport departments, the police, traffic cameras and road sensors [1]. The authors also added the weather from the closest airport station and nearby road features from OpenStreetMap [2]. We use the random sample of 500,000 rows that is in our repository (US_Accidents_March23_sampled_500k.csv). It is one file, so we did not need to merge anything.

2.2 Size and structure

The sample has 500,000 rows (one row = one accident) and 46 columns: 20 text, 13 True/False, 12 decimal numbers and 1 whole number. Table 2 puts them into groups. Appendix A explains every column, with its type, unit and how much of it is missing.

Table 2. Columns of the raw sample. The units are written in the column names, for example (F) = degrees Fahrenheit.

Group	Columns	Type and unit
Target	Severity	Whole number 1-4 (effect on traffic)
Identifiers	ID, Source	Text
Time	Start_Time, End_Time, Weather_Timestamp, Timezone	Date-time (local time), text
Location	Start_Lat, Start_Lng, End_Lat, End_Lng, Distance(mi)	Degrees; distance in miles
Address	Street, City, County, State, Zipcode, Country, Airport_Code	Text
Description	Description	Free text written by the provider
Weather	Temperature(F), Wind_Chill(F), Humidity(%), Pressure(in), Visibility(mi), Wind_Speed(mph), Precipitation(in), Wind_Direction, Weather_Condition	°F, %, inches of mercury, miles, mph, inches; text
Road features	Amenity, Bump, Crossing, Give_Way, Junction, No_Exit, Railway, Roundabout, Station, Stop, Traffic_Calming, Traffic_Signal, Turning_Loop	True / False (is this near the accident?)
Light	Sunrise_Sunset, Civil_Twilight, Nautical_Twilight, Astronomical_Twilight	Day / Night

Table 3. The target variable. The sample has almost the same class shares as the full dataset (less than 0.1 percentage point difference).

Severity	Meaning	Accidents (sample)	Share (sample)	Share (full 7.7M)
1	Smallest impact	4,274	0.85%	0.87%
2	Medium	398,142	79.63%	79.67%
3	Big	84,520	16.90%	16.81%
4	Biggest impact (long delay)	13,064	2.61%	2.65%

Key variables. The target is Severity. Based on our question, the most important inputs are the time (Start_Time), the place (coordinates, state, city, street), the weather columns and the 13 road features.

2.3 First data quality problems

Table 4. Data quality problems we found when we first looked at the data.

Problem	What we found	Fixed in
Missing values	End_Lat/End_Lng 44%, Precipitation 29%, Wind_Chill 26%, Wind_Speed 7.4%, other weather about 2%, light columns 0.3%, City/Zipcode less than 0.03%	3.3
Leakage columns	End_Time, End_Lat/Lng, Distance(mi) and Description are only known after the accident	3.1
Impossible values	Temperature of 207 °F and -77.8 °F, wind speed of 822.8 mph, visibility of 130 miles, pressure of 0.12 inHg	3.4
Messy categories	Wind_Direction is written in 24 ways (for example "West" and "W", "Calm" and "CALM"); Weather_Condition has 108 labels	3.5
Mixed date formats	Start_Time is written both with and without fractions of a second	3.2
Constant columns	Country is always "US" and Turning_Loop is always False	3.1
Duplicates	None when the ID is included; 2,884 identical rows once the ID is removed	3.2
Provider differences	The three providers use the Severity levels in very different ways (Section 4.4)	4.4, 5

Limits of the data. It is only a sample (6.5% of the full dataset). It stops in March 2023, so 2023 has only 3 months. How much data there is depends on the providers, so some states and years have more data than others. Severity is given by the providers, and they do not all measure it the same way.

3. Data Pre-processing

All the steps are in the notebook 02_cleaning.ipynb, and each step is written down in reports/cleaning_log.md. The data went from 500,000 × 46 to 495,532 × 36.

3.1 Stopping data leakage: choosing columns

Our rule: we only keep a column if its value is known when the accident is first reported. If a model learns from information that comes later, it looks great in testing but is useless in real life. Table 5 shows that these "later" columns basically contain the answer.

Table 5. Proof of leakage. For each Severity level: the median time the accident lasted, the average length of road affected, and the share of descriptions that say "closed".

Severity	Median duration (End - Start)	Average road affected	Description says "closed"
1	45 min	0.11 mi	2.2%
2	78 min	0.57 mi	4.7%
3	44 min	0.42 mi	5.4%
4	129 min	1.48 mi	88.5%

These columns separate Severity 4 from the others almost perfectly, so we must remove them. Table 6 lists all removed columns.

Table 6. The 11 removed columns. We kept Source on purpose (we will test the model with and without it). We also kept City, County, Street and Zipcode; they have thousands of values, so we handle them in feature engineering.

Removed column	Reason
End_Time	Leakage: only known once the road is clear (a long duration means a severe accident)
End_Lat, End_Lng	Leakage: the end of the affected road, measured afterwards
Distance(mi)	Leakage: the length of road affected is part of how Severity is decided
Description	Leakage: has text like "road closed" or "lanes blocked"
ID	Just an identifier, no useful information
Country, Turning_Loop	Constant: the same value in every row (we checked)
Weather_Timestamp, Timezone, Airport_Code	Repeats what Start_Time and the location columns already tell us

We also avoid leakage in later steps. Anything that learns from the data (filling with the median, encodings, clusters, scaling, feature selection) is fitted on the training data only (Section 5).

3.2 Types and duplicates

- We changed Start_Time into a date-time with format="mixed", so both date formats work.
- We changed the 12 road-feature columns from True/False to 1/0, so models can use them.
- We removed 2,884 rows that were exact copies. Otherwise the same accident would count twice.

3.3 Missing values

We checked how many cells are empty in each column. Then we picked a fix for each case (Table 7).

Table 7. How we handled missing values. Percentages are the share of rows in the sample.

Column(s)	Missing	What we did	Why
Wind_Chill(F)	26%	Removed the column	It is almost a copy of Temperature (correlation 0.994)
Precipitation(in)	29%	Filled with 0, and added a new column precip_missing (1 = was empty)	Empty mostly means no rain was reported. The new column remembers that we guessed.
City, Zipcode, light columns	0.3%	Removed the rows (1,584 rows)	Too few rows to be worth guessing
Weather_Condition, Wind_Direction	2.2%	Filled with "Unknown"	We keep the rows, and "Unknown" becomes its own category
Temperature, Humidity, Pressure, Visibility, Wind_Speed	2-7%	Left empty for now; filled with the training median when the model is trained	Filling now would use test data (leakage). The median is not pulled by extreme values.

3.4 Outliers and impossible values

We checked the value ranges with summary statistics and box plots (Figure 1). Values that cannot exist in real life were set to empty, and then treated like other missing values. Values that are extreme but possible were kept.

Table 8. Limits for impossible values.

Column	Allowed range	Values removed	Why this limit
Temperature	-50 to 130 °F	4	The US record high is 134 °F; below -50 °F is not realistic on these roads
Wind speed	0 to 200 mph	3	Stronger than any wind measured on the ground outside tornadoes
Visibility	0 to 100 miles	3	Weather stations do not report more than about 100 miles
Pressure	15 to 32 inHg	3	The highest US roads are about 17.5 inHg; the world record high is 32.06 inHg

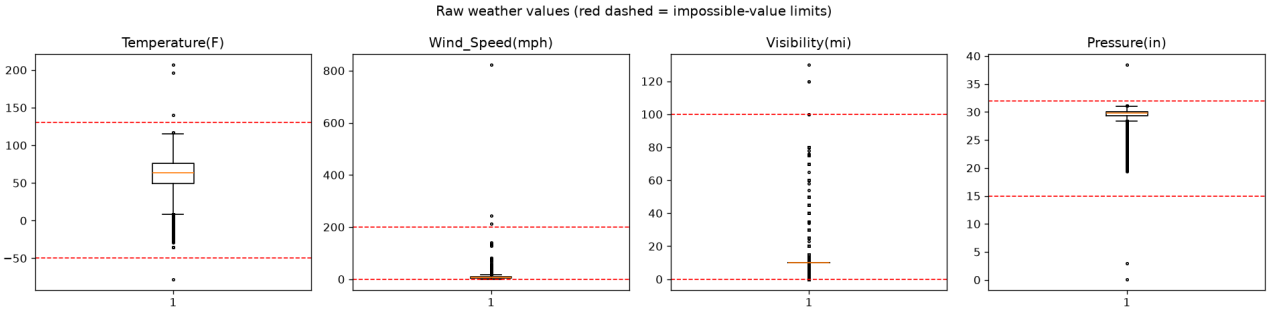


Figure 1. Weather values before cleaning. Box plots of the raw values. The red dashed lines are the limits from Table 8.

3.5 Messy categories

- **Wind_Direction:** we merged the 24 spellings into 16 compass points + CALM + VAR (+ Unknown). For example "West" → "W", "Calm" → "CALM", "Variable" → "VAR".
- **Weather_Condition:** we grouped the 108 labels into 8 groups + Unknown, in a new column called Weather_Group: Clear, Cloudy, Rain, Heavy rain / Thunderstorm, Snow / Ice, Fog / Haze / Smoke, Windy, Other. The rules go from most to least dangerous, so "Light Snow with Thunder" counts as Thunderstorm.

3.6 Effect on the dataset

Table 9. Size of the dataset after each step. After cleaning, only 5 weather columns (2-7%) and Street (0.1%) still have empty cells.

Step	Rows	Columns
Raw sample	500,000	46
After removing leaking / useless columns	500,000	35
After removing duplicates	497,116	35
After fixing missing values	495,532	35
After grouping the weather (final clean data)	495,532	36

In total we removed only 0.9% of the rows.

4. Exploratory Data Analysis

All charts and tests come from the notebook 03_eda.ipynb (the images are in reports/figures/). "High impact" means Severity 3 or 4. Section 4.1 describes the variables. Section 4.2 shows when and where accidents happen. Section 4.3 links the variables to Severity (RQ1-RQ3). Section 4.4 checks if the label is stable (RQ4). Section 4.5 tests the main links with statistics.

4.1 Descriptive statistics

Table 10. Summary of the numeric weather columns (clean data). A skew above 0 means a long tail of high values; below 0 means a long tail of low values.

Variable	Mean	Median	Std	Min	Max	Missing	Skew
Temperature(F)	61.64	64.00	18.99	-35	117	2.0%	-0.51
Humidity(%)	64.84	67.00	22.83	1	100	2.2%	-0.40
Pressure(in)	29.54	29.86	1.01	19.36	31.15	1.7%	-3.68
Visibility(mi)	9.09	10.00	2.69	0	100	2.2%	2.32
Wind_Speed(mph)	7.68	7.00	5.27	0	142	7.4%	0.93
Precipitation(in)	0.01	0.00	0.09	0	10.13	0.0%	91.80

Table 10 shows that most accidents happen in mild, dry and clear weather (median 64 °F, 10 miles of visibility, no rain). Precipitation (skew 92) and visibility are very skewed. Pressure is skewed the other way because some stations are high up. For these columns the median describes the data better than the mean. That is also why we fill missing values with the median. For the text columns: Source1 reports 56% of the accidents, California has 23% and "Clear" weather is 44%.

Figure 2 shows the target. **What it shows:** the classes are very unbalanced. 80% are Severity 2, only 0.9% are Severity 1 and 2.6% are Severity 4. A model that always says "2" would be 80% accurate but useless. That is why we will not use accuracy, and use macro F1 instead.

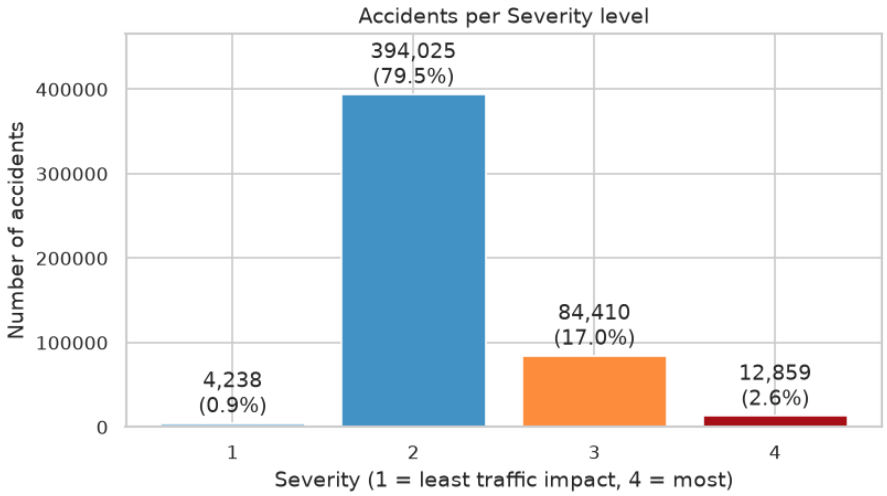


Figure 2. Number of accidents per Severity level. The bar height is the count; the labels give the share of all accidents.

4.2 When and where accidents happen

Before we look at Severity, Figures 3-5 show how the accidents are spread over time and over the map.

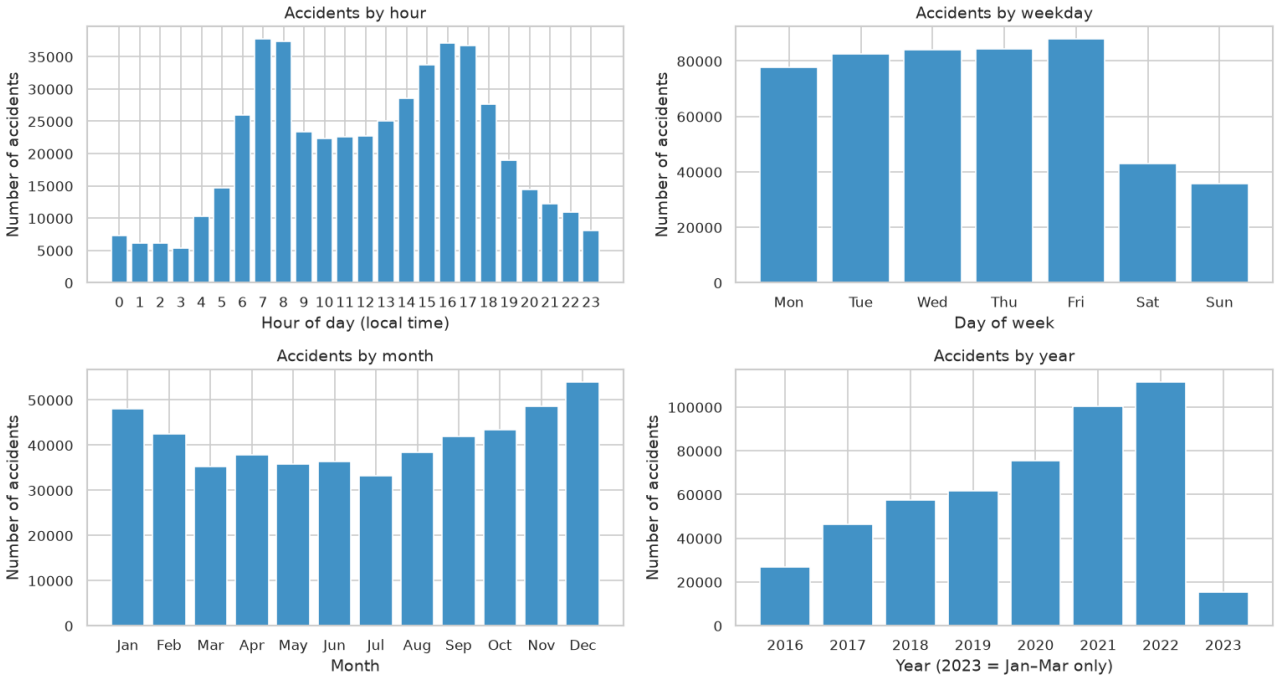


Figure 3. Number of accidents by hour, weekday, month and year. Hours are in local time. 2023 only has January to March.

What Figure 3 shows: accidents follow the times people go to and from work. There are two peaks, at 7-8 h and at 15-17 h. On Saturdays and Sundays there are about half as many accidents as on weekdays. December and January are the busiest months (dark evenings, winter weather, holidays). The number per year goes up from 26,700 (2016) to 113,700 (2022). This is mostly because the data collection got better, not because there were more accidents. It is a first sign that the data changes over time (Section 4.4).

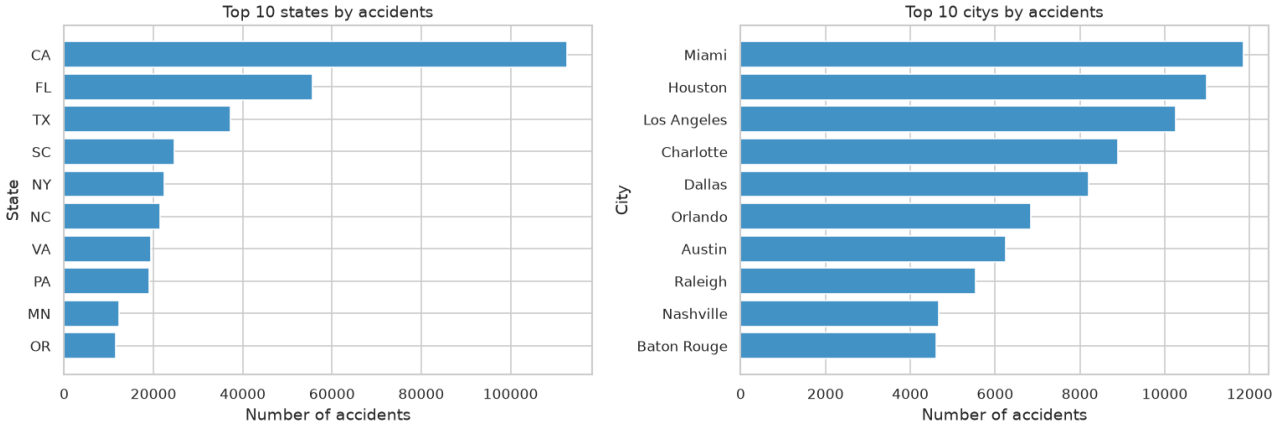


Figure 4. The top 10 states and cities by number of accidents. Horizontal bars; counts in the clean sample.

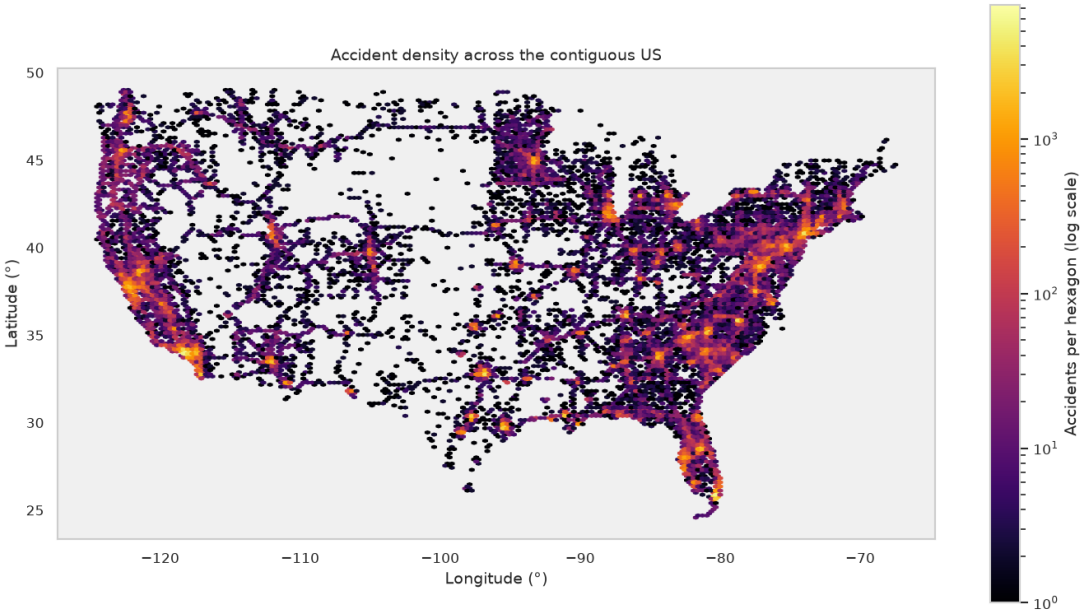


Figure 5. Where accidents happen in the mainland United States. Each hexagon counts the accidents inside it; the colour scale is logarithmic.

What Figures 4-5 show: California alone has 23% of the accidents, then Florida (11%) and Texas (7.5%). The top cities are Miami, Houston and Los Angeles. On the map, accidents follow the big highways and large cities. The Mountain West is almost empty. The counts depend on how many people live there and on how well each provider covers the area. So location is a strong signal, but it is partly biased.

4.3 Links with Severity (RQ1-RQ3)

RQ1 (time). Figure 6 shows the Severity mix for each hour, and Figure 7 compares day and night.

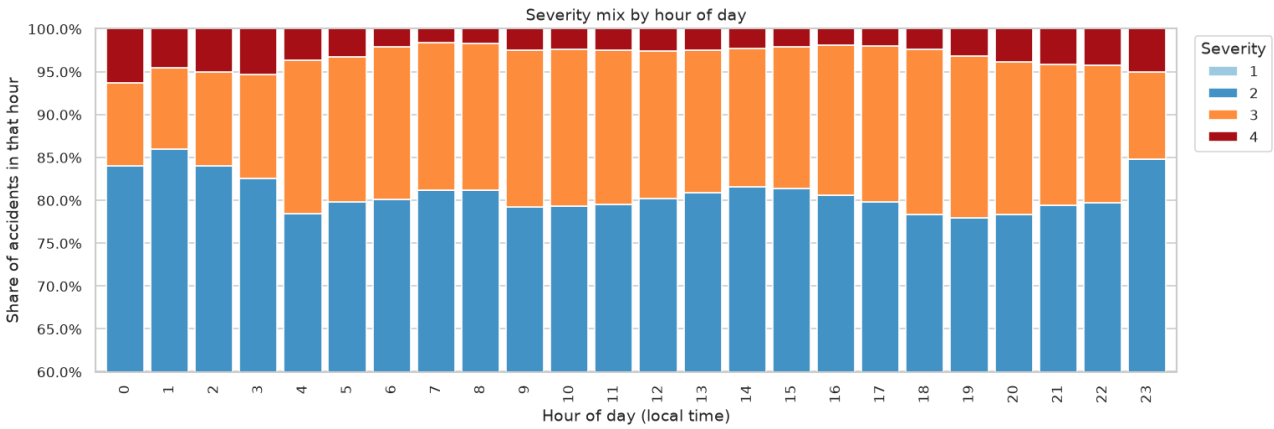


Figure 6. Severity mix by hour of the day. Each bar is one hour (local time) and adds up to 100%. The y-axis starts at 60% so the small classes are easier to see.

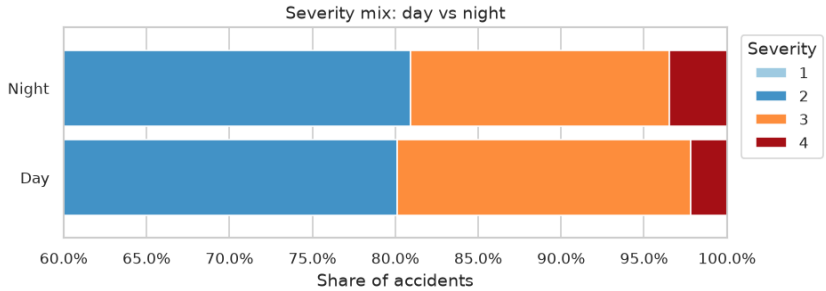


Figure 7. Severity mix by day and night. Day and night are based on sunrise and sunset. The x-axis starts at 60%.

What it shows: rush hours have the most accidents, but not the most severe ones. Severity 4 is lowest at 7-8 h (1.6%) and highest around midnight (5-6%). At night, Severity 4 is 1.6 times more common than during the day (3.5% vs 2.2%). Severity 3 is a bit less common at night (15.6% vs 17.7%). Possible reasons: people drive faster on empty highways, it is harder to see, and help arrives more slowly at night.

RQ2 (weather). Figure 8 compares the weather groups, and Figure 9 compares visibility and temperature across Severity levels.

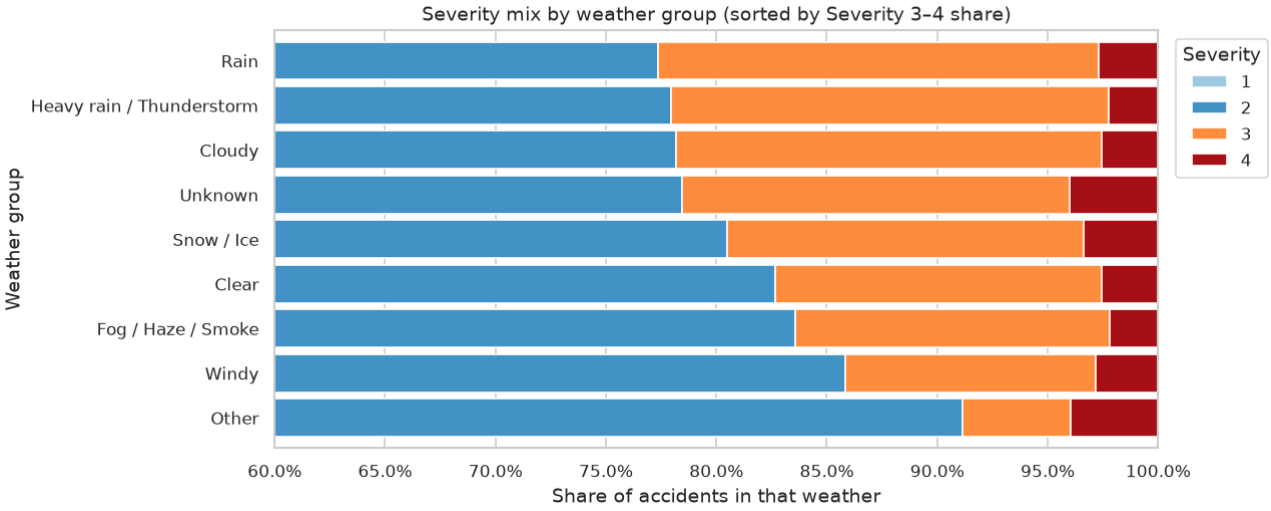


Figure 8. Severity mix by weather group. Groups are sorted by their share of Severity 3-4. The x-axis starts at 60%.

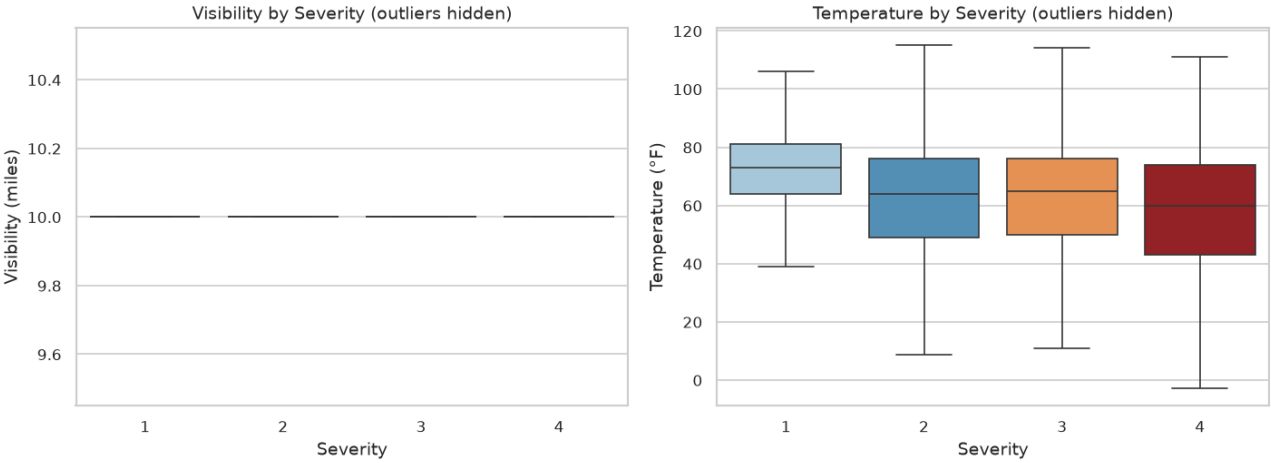


Figure 9. Visibility and temperature by Severity level. Box plots without outliers: the box holds the middle 50% of accidents and the line is the median.

What it shows: in rain and thunderstorms about 22% of accidents are high-impact, compared to 17% in clear weather. Snow and ice make Severity 4 more common (3.4% vs 2.6%). Visibility is the same for every level (median 10 miles, because most accidents happen when you can see well). Severity 4 accidents happen at slightly lower temperatures (median 60 °F vs 64 °F for Severity 2). So the weather matters, but much less than we expected.

RQ3 (road). Figure 10 compares accidents with and without four common road features.

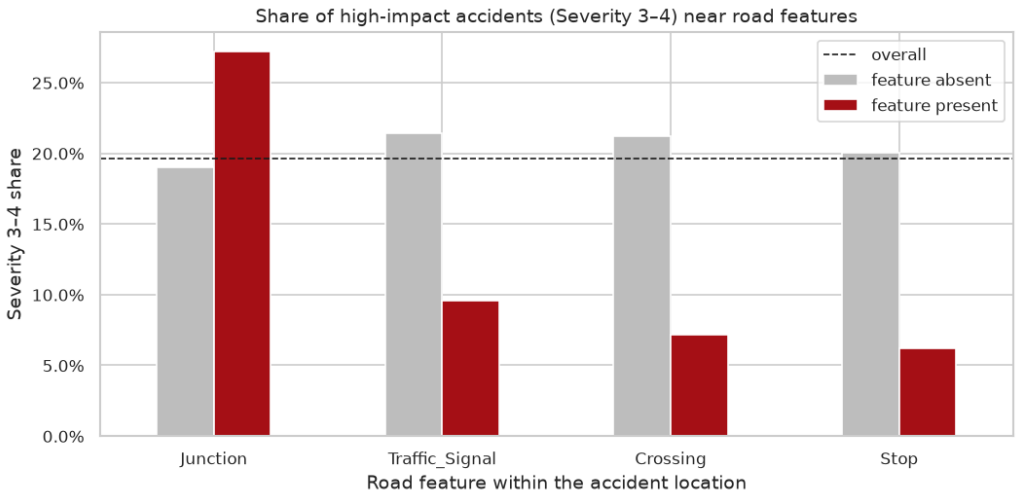


Figure 10. Share of high-impact accidents near road features. Grey: feature not there; red: feature there; dashed line: overall share (19.6%).

What it shows: accidents near traffic lights, crossings or stop signs are 2-3 times less often high-impact (6-10% vs about 20%). This is because these are slow city streets. Junctions are different (27%), because they include highway exits. In Section 5 we find that highway accidents are 28% high-impact, compared to 11% on other roads. The type of road is one of the clearest signals in the data.

Correlations. Figure 11 shows the Pearson correlations between the numeric columns.

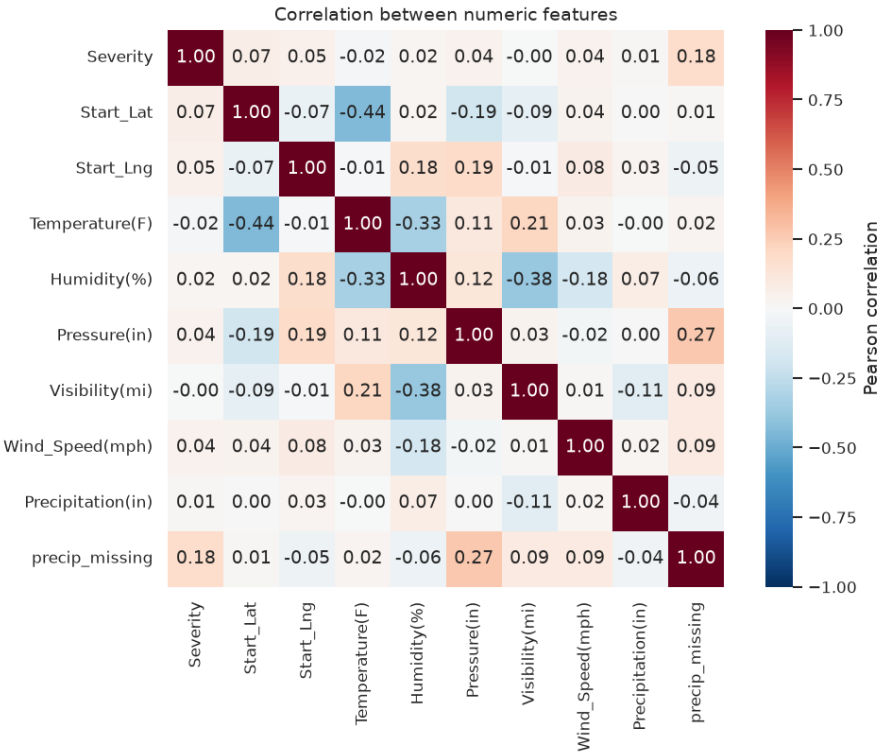


Figure 11. Correlation between numeric features. Pearson correlation; red = positive, blue = negative.

What it shows: no numeric column has a straight-line link with Severity (all |r| are below 0.1). The only clear pairs are temperature with latitude (-0.44: the north is colder), and humidity with temperature (-0.33) and with visibility (-0.38: fog and rain). No pair is strong enough to make a column useless. So the signal is not a straight line and comes from combinations of columns. This points to tree-based models later.

4.4 Changes over time and between providers (RQ4)

Figure 12 shows the Severity mix per year, and Figure 13 shows the role of the three data providers.

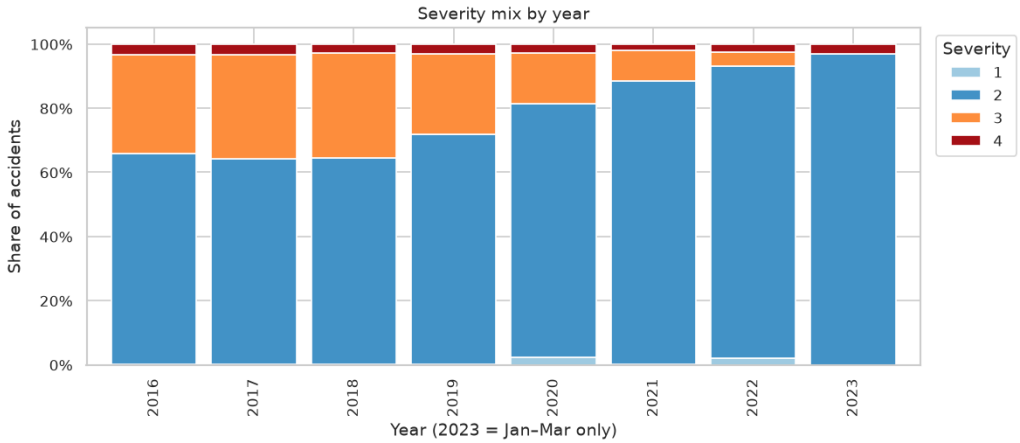


Figure 12. Severity mix by year. Each bar adds up to 100%. 2023 = January to March only.

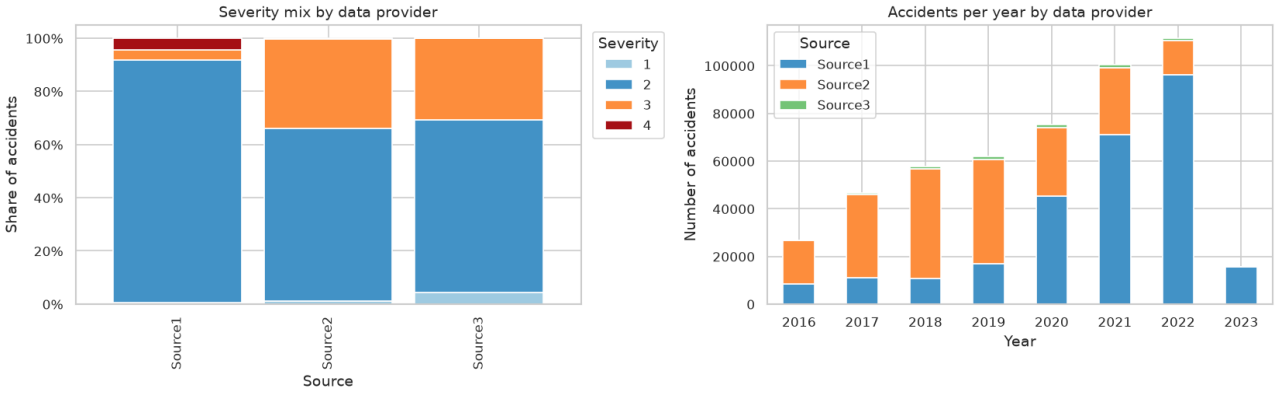


Figure 13. Severity by data provider (left) and accidents per year by provider (right). Left: each bar adds up to 100%. Right: stacked counts per year.

What it shows (RQ4): the label is **not** stable. Severity 3 drops from about 32% of accidents (2017-2018) to 4.5% (2022) and 0% (2023). Severity 1 shows up almost only in 2020 and 2022. Figure 13 explains why. Source2 calls 34% of its accidents Severity 3, but Source1 only 4%. On the other hand, Source1 calls 4.3% of its accidents Severity 4, against 0.4% for Source2. Over the years, Source1 replaced Source2. So part of "severity" shows who recorded the accident, not only the accident itself. This matters for the model: it might learn the provider instead of the accident, and scores on recent years will be lower.

4.5 Statistical tests

A chart can show a pattern that is just luck, so we tested the patterns (significance level $\alpha = 0.05$). For two categorical variables we used the **chi-square test of independence**. H0 says the variable and Severity are not linked. This test needs every expected count to be at least 5, so we left out the 203 rows with "Other" weather. For temperature across the 4 levels we used the **Kruskal-Wallis test** (H0: the same distribution in every group) instead of ANOVA. We did this because temperature is not normally distributed (D'Agostino test, $p \approx 10^{-41}$) and the groups do not have the same variance (Levene test, $p \approx 10^{-194}$).

With almost 500,000 rows, even tiny differences give $p \approx 0$. So we also report the **effect size**, which tells how strong the link is (Cramér's V: 0.1 small, 0.3 medium, 0.5 large; ε^2: 0.01 small, 0.06 medium). We also ran each test again on a random sample of 5,000 rows.

Table 11. Results of the statistical tests. "≈ 0" means smaller than 10^{-300}. The effect size shows how strong the link is.

H0: Severity is not linked to...	Test	Statistic	p-value	p-value (5,000 rows)	Effect size	Decision
Data provider (Source)	Chi-square	χ^2 = 82,258	≈ 0	4×10^{-182}	V = 0.29 (small, close to medium)	Reject H0
Traffic signal nearby	Chi-square	χ^2 = 7,336	≈ 0	8×10^{-13}	V = 0.12 (small)	Reject H0
Day / night	Chi-square	χ^2 = 1,242	7×10^{-269}	0.0004	V = 0.05 (very small)	Reject H0
Junction nearby	Chi-square	χ^2 = 1,466	≈ 0	0.13	V = 0.05 (very small)	Reject H0 (weak)
Weather group	Chi-square	χ^2 = 2,264	≈ 0	0.013	V = 0.04 (very small)	Reject H0
Temperature (same for all levels)	Kruskal-Wallis	H = 1,818	≈ 0	2×10^{-5}	ε^2 = 0.004 (very small)	Reject H0

What it shows: we reject every H0 on the full data, but the effect sizes tell us what really matters. The data provider has by far the strongest link with Severity (this confirms RQ4). Next comes the road type (traffic lights). Light, weather and junctions have very small effects. They are real but weak, and the junction effect is not even significant on 5,000 rows. Temperature is only a little different between levels. **Conclusion:** no single column explains Severity, so a model will need to combine many weak signals.

5. Feature Engineering

This is done in the notebook 04_features.ipynb, with reusable code in src/features.py. There are two kinds of steps. Some features are built from one row only, so they are safe to build at any time. Other steps learn from many rows, so they are fitted on the training data only.

5.1 New features

Table 12. Features built from each row. Last column: share of Severity 3-4 when the feature is 1 vs 0.

New feature	How it is built	Why	High-impact share (yes vs no)
hour, weekday, month, year	From Start_Time	Traffic and reporting change with time	-
season	Month → Winter / Spring / Summer / Autumn	Snow and ice in winter	-
is_weekend	1 if Saturday or Sunday	Different traffic	22% vs 19%
is_rush_hour	1 if weekday and hour is 7-9 or 16-19	Traffic jams	20% vs 20%
civil_night, astro_night	1 if dark (civil / astronomical twilight)	Darkness, fewer helpers	19% vs 20%
weather_group, bad_weather	9 weather groups; 1 if rain, storm, snow/ice or fog	Fewer, stronger categories	21% vs 19%
road_feature_count	Number of road features nearby (0-12)	City street vs open road	22% (0) → 16% (1) → 8% (2+)
is_highway	Street name matches I-95, US-101, CA-99, Hwy, Fwy, Expy, Tpke, Beltway, Route...	Highway accidents block more traffic	28% vs 11%
precip_missing	1 if precipitation was empty (Section 3.3)	Remembers that we guessed the value	-

is_highway is the strongest new signal. We checked the highway rule on the most common street names. The first version missed state roads like "CA-99" and "Capital Beltway". After we fixed it, the streets that do not match are local ones like "Main St".

5.2 Splitting the data before fitting anything

Before choosing or encoding anything, we split the data into 80% training (396,425 rows) and 20% test (99,107 rows). Both parts have the same Severity shares (stratified split, seed 42). Every decision below is made on the training part only. This way the test part stays unseen, just like new data for a real model.

5.3 Choosing features (RQ5)

We started with 41 possible inputs and did three checks on the training data:

- **Almost constant columns:** a yes/no column that is 1 in less than 0.1% of rows tells us almost nothing. So we removed Roundabout (0.002%), Bump (0.04%) and Traffic_Calming (0.09%). They are still counted in road_feature_count, so we do not lose them completely.
- **Repeated columns:** when two columns are almost the same (|correlation| above 0.85), we keep the one that tells us more about Severity. So we removed is_night (r = 0.89 with civil_night) and nautical_night (r = 0.88 with astro_night).
- **Mutual information (MI):** this measures how much a feature tells us about Severity. It also works for non-linear links and for categories. We used it to rank the features (Figure 14) and to choose between repeated columns.

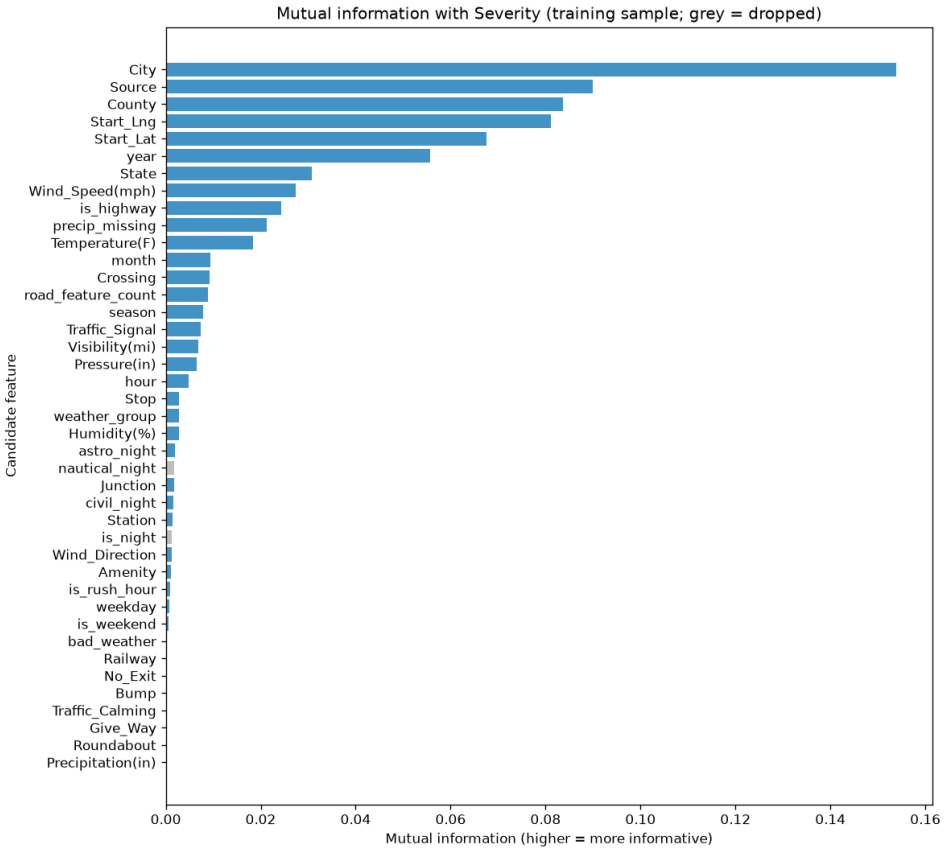


Figure 14. Mutual information between each possible feature and Severity. Computed on 60,000 training rows; grey bars are the removed features.

What it shows (RQ5): location (City, County, coordinates, State), the data provider (Source), the year, wind speed and is_highway tell us the most. City and County look a bit better than they really are, because MI tends to give higher scores to columns with thousands of values. We keep weak features like the weather. The EDA showed they matter a little, and tree models can combine them.

5.4 Encoding and scaling

Table 13. Steps fitted on the training data only. They are all put together in one scikit-learn ColumnTransformer (src/features.py).

Step	Columns	What it does	Why we chose it
Median filling	Numeric columns	Fills empty cells with the training median	Not pulled by the extreme weather values
One-hot encoding	State, Source, weather_group, Wind_Direction, season	One 0/1 column per value (84 columns)	Few values (49 at most), and no fake order between them
Frequency encoding	City, County	Replaces each name with its share of training rows; unseen names → 0	One-hot would make about 10,000 mostly empty columns
Location clusters	Start_Lat, Start_Lng	K-means splits the map into 50 regions, then one-hot	Finds regional hotspots that raw coordinates show badly (Figure 15)
Standard scaling	Numeric columns	Mean 0, standard deviation 1	Only for linear or distance-based models; trees do not need it

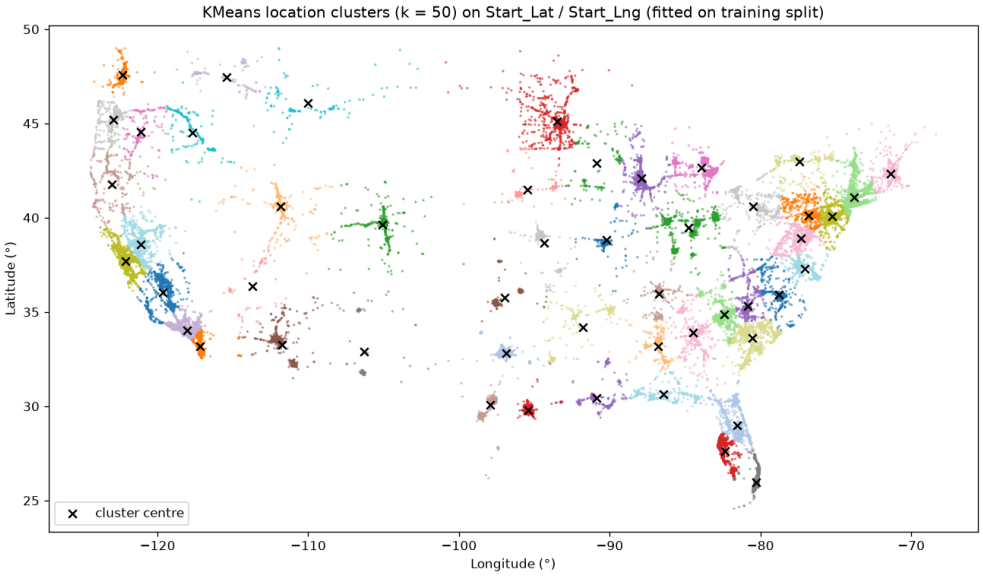


Figure 15. The 50 location clusters. K-means on the latitude and longitude of the training accidents; the crosses are the cluster centres.

5.5 Final feature set and checks

Table 14. Final feature set. Saved in data/features.parquet (495,532 rows × 36 inputs + Severity).

Group	Final inputs (36)	Why we kept them
Location	Start_Lat, Start_Lng, State, City, County	Most informative; Severity changes by region
Road	is_highway, road_feature_count, Junction, Traffic_Signal, Crossing, Stop, Station, Railway, Give_Way, No_Exit, Amenity	Road type has the biggest real effect (RQ3)
Time	hour, weekday, month, year, season, is_weekend, is_rush_hour	Time patterns (RQ1); year captures changes in reporting
Light	civil_night, astro_night	Night makes Severity 4 more common (RQ1)
Weather	Temperature, Humidity, Pressure, Visibility, Wind_Speed, Wind_Direction, Precipitation, precip_missing, weather_group, bad_weather	Small effects that may add up (RQ2)
Provider	Source	Strongest link (RQ4); we will test the model with and without it

How the features help our goal. Every input is known when the accident is reported, so the model can be used at that moment. That is the goal from Section 1. The new features turn raw columns into the patterns we found in the EDA (night, road type, bad weather, rush hour). The selection removes columns that add noise but no information. The encodings turn text into numbers without making thousands of empty columns and without leaking test data. We kept Source on purpose, so we can measure its effect when we build the model (RQ4).

Checks: after fitting on the training data and applying it to the test data, both have 165 numeric columns (29 numeric, 84 one-hot, 2 frequency, 50 regions) and no missing values. 399 test accidents (0.4%) come from cities that are not in the training data, and they get frequency 0. This proves that nothing was learned from the test data. After scaling, the training columns have mean 0 and standard deviation 1, and the test columns are very close (within 0.01), as expected.

6. Conclusions and Next Steps

6.1 Answers to the research questions

Table 15. Answers to the research questions.

Question	Answer (with evidence)
RQ1 Time	Rush hour has more accidents, but not worse ones. Severe (level 4) accidents are most common at night: 5-6% around midnight vs 1.6% at 7-8 h; 3.5% at night vs 2.2% during the day (small effect, V = 0.05).
RQ2 Weather	Small effect: rain and storms raise high-impact accidents from 17% to about 22%, and snow/ice raises Severity 4. It is statistically significant but very small (V = 0.04). Temperature is only a little different.
RQ3 Road	Clear effect: highway accidents are 28% high-impact vs 11% on other roads. Near traffic lights, crossings and stop signs it is only 6-10% (V = 0.12 for traffic lights).
RQ4 Consistency	No. The providers label Severity very differently (V = 0.29, the strongest link), and the mix of providers changes over the years, so Severity changes over time.
RQ5 Features	Location, provider, year and highway tell us the most. We kept 36 inputs after removing 3 almost constant and 2 repeated columns.

6.2 Main findings

- The target is very unbalanced (80% Severity 2), so future models need macro F1 and class balancing.
- Leakage was a real danger: 88.5% of Severity 4 descriptions say "closed". Removing these columns makes the task honest, but harder.
- Severity depends more on **where** (road type, region) and **who reported it** than on the weather or the time.
- No single column is strongly linked to Severity. The signal is weak and not a straight line, so tree-based models should do better than linear ones.

6.3 Limitations

- Severity measures traffic delay as recorded by each provider, not injuries, and the providers use the scale differently. A model might partly learn the provider's habits instead of the real impact.
- We used a 6.5% sample. The data stops in March 2023, and coverage changes by state and year.
- The weather comes from the closest airport station, so it may not match the exact place of the accident.
- Some choices are rules of thumb: the limits for impossible values, the highway name rule, the 0.1% and 0.85 thresholds, and filling missing precipitation with 0.
- The statistical tests show a link, not a cause.

6.4 Next steps (building the model, not part of this midterm)

- **Splits:** split the training part again into train and validation (70/15/15 overall) for all choices. Keep the test set for one final check. Also do a time-based check (train on 2016-2021, test on 2022-2023).
- **Unbalanced classes:** on the validation data, compare class weights with removing some Severity 2 rows (undersampling).
- **Models:** a lazy baseline (always Severity 2, macro F1 about 0.22) → Logistic Regression → Decision Tree → Random Forest → LightGBM / XGBoost, all using the preprocessing from Section 5.
- **Evaluation:** macro F1, recall of Severity 4, balanced accuracy and a confusion matrix.
- **Provider test:** train the best model with and without Source and report the difference.
- **Explanation:** use feature importance and SHAP values to see which features drive the predictions.

References

[1] Moosavi, S., Samavatian, M. H., Parthasarathy, S., & Ramnath, R. (2019). A Countrywide Traffic Accident Dataset. arXiv:1906.05409.

[2] Moosavi, S., Samavatian, M. H., Parthasarathy, S., Teodorescu, R., & Ramnath, R. (2019). Accident Risk Prediction based on Heterogeneous Sparse Data: New Dataset and Insights. Proceedings of the 27th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems.

[3] Moosavi, S. US Accidents (2016-2023) [Dataset]. Kaggle. https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents

[4] Moosavi, S. US-Accidents: A Countrywide Traffic Accident Dataset. https://smoosavi.org/datasets/us_accidents

[5] Software: pandas, NumPy, SciPy, scikit-learn, matplotlib and seaborn (Python 3.12).

All figures and tables were made by our team from the dataset [1-3]. The dataset is used under its CC BY-NC-SA 4.0 licence.

Appendix A: What each column means

Table 16. Meaning of every column in the raw sample. Type = how pandas loads it; Missing = share of empty cells in the raw sample.

Column	Type	Unit	Meaning	Missing
ID	text	-	Unique code of the accident record	0.0%
Source	text	-	Data provider that reported the accident (Source1, Source2, Source3)	0.0%
Severity	integer	1-4	Effect on traffic: 1 = short delay, 4 = long delay (target)	0.0%
Start_Time	text	local time	When the accident started	0.0%
End_Time	text	local time	When its effect on traffic ended (only known afterwards)	0.0%
Start_Lat	decimal	degrees	Latitude of the start point	0.0%
Start_Lng	decimal	degrees	Longitude of the start point	0.0%
End_Lat	decimal	degrees	Latitude of the end of the affected road	44.1%
End_Lng	decimal	degrees	Longitude of the end of the affected road	44.1%
Distance(mi)	decimal	miles	Length of road affected by the accident	0.0%
Description	text	text	Free-text description written by the provider	0.0%
Street	text	-	Street name	0.1%
City	text	-	City	0.0%
County	text	-	County	0.0%
State	text	-	US state (2 letters)	0.0%
Zipcode	text	-	Postal code	0.0%
Country	text	-	Country (always US)	0.0%
Timezone	text	-	Time zone of the place	0.1%
Airport_Code	text	-	Closest airport weather station	0.3%
Weather_Timestamp	text	local time	Time of the weather reading	1.5%
Temperature(F)	decimal	°F	Air temperature	2.1%
Wind_Chill(F)	decimal	°F	How cold it feels with the wind	25.8%
Humidity(%)	decimal	%	Relative humidity	2.2%
Pressure(in)	decimal	inHg	Air pressure	1.8%
Visibility(mi)	decimal	miles	How far you can see	2.3%
Wind_Direction	text	-	Wind direction (compass point, CALM or VAR)	2.2%
Wind_Speed(mph)	decimal	mph	Wind speed	7.4%
Precipitation(in)	decimal	inches	Amount of rain or snow	28.5%
Weather_Condition	text	-	Weather description (rain, snow, fog, ...)	2.2%
Amenity	True/False	True/False	A shop, school or similar place near the accident	0.0%
Bump	True/False	True/False	Speed bump nearby	0.0%
Crossing	True/False	True/False	Pedestrian crossing nearby	0.0%
Give_Way	True/False	True/False	Yield sign nearby	0.0%
Junction	True/False	True/False	Junction or highway exit nearby	0.0%
No_Exit	True/False	True/False	Dead-end road nearby	0.0%

Column	Type	Unit	Meaning	Missing
Railway	True/False	True/False	Railway nearby	0.0%
Roundabout	True/False	True/False	Roundabout nearby	0.0%
Station	True/False	True/False	Train or bus station nearby	0.0%
Stop	True/False	True/False	Stop sign nearby	0.0%
Traffic_Calming	True/False	True/False	Road design that slows cars down nearby	0.0%
Traffic_Signal	True/False	True/False	Traffic lights nearby	0.0%
Turning_Loop	True/False	True/False	Turning loop nearby	0.0%
Sunrise_Sunset	text	Day/Night	Day or night, based on sunrise and sunset	0.3%
Civil_Twilight	text	Day/Night	Day or night, based on civil twilight (sun 6° below the horizon)	0.3%
Nautical_Twilight	text	Day/Night	Day or night, based on nautical twilight (12°)	0.3%
Astronomical_Twilight	text	Day/Night	Day or night, based on astronomical twilight (18°)	0.3%

Appendix B: How to run it again

Install the packages in requirements.txt, then run **python run_all.py** from the project folder. It runs notebooks 01 → 04 in order (about one minute) and makes every file used in this report again: data/sample.parquet, data/clean.parquet, data/features.parquet, reports/leakage_audit.md, reports/cleaning_log.md, reports/data_dictionary.md and reports/figures/. See README.md for setup details.